



ME 300:
Engineering Thermodynamics

Non-Ideal Gases

Non-ideal gases

Ideal gases $\rightarrow V_{\text{mol}} \ll V_{\text{b/w mol}}$
 $\rightarrow$ no intermolecular forces

Non-ideals $\rightarrow$ density is high, but T isn't really
 $\rightarrow$ intermolecular forces are important

Van der Waals equation of state

$$PV = NR_uT \quad \sim \bar{V} \sim \left[\frac{\text{m}^3}{\text{kmol}} \right]$$
$$P\bar{V} = R_uT$$

$$\Rightarrow \left(P + \frac{a}{\bar{V}^2} \right) (\bar{V} - b) = R_uT$$

a - accounts for intermolecular forces

b - accounts for volume of mol.

Example – Van der Waals EOS

$$\text{CH}_4 \rightarrow M = 16.04 \text{ kg/kmol} \rightarrow R = \frac{R_u}{M} = 518.4 \frac{\text{J}}{\text{kg K}}$$

$$T = 300 \text{ K}$$

$$\rho = 175 \text{ kg/m}^3$$

$$\rightarrow \text{ideal gas: } P = \rho R T = (175)(518.4)(300) \quad P_{\text{ideal gas}} = \boxed{27.2 \text{ MPa}}$$

$$\rightarrow \text{V.d.W.: } v = \frac{1}{\rho} = 0.0057 \text{ m}^3/\text{kg}, \quad \bar{v} = v M = 0.0917 \text{ m}^3/\text{kmol}$$

$$\left(P + \frac{a}{\bar{v}^2}\right)(\bar{v} - b) = R_u T$$

$$P = \frac{R_u T}{\bar{v} - b} - \frac{a}{\bar{v}^2} = \frac{(8314.5)(300)}{(0.0917 - 0.0427)} - \frac{2.285 \times 10^5}{(0.0917)^2} = \boxed{23.7 \text{ MPa}}$$